Title: Conversion of ER diagram into Relational Schema and implementation using MySQL.

Aim: To convert the given E-commerce ER diagram into a relational diagram and implement it in MySQL using Primary Keys, Foreign Keys, NOT NULL, UNIQUE constraints, ON DELETE CASCADE and ON DELETE SET NULL. To insert sample data and demonstrate referential integrity violations.

Theory

An Entity-Relationship (ER) Model represents the entities, attributes and relationships of a database system. To implement an ER model in a relational database, entities are converted into tables and relationships are represented using primary and foreign keys.

Rules used for conversions:

1. Each strong entity is converted into a separate solutions.
2. The primary key of an entity becomes the primary key of its relation.
3. A composite attribute is divided into its simple attributes.
4. A multivalued attribute is represented using a separate relation.
5. A 1:N relationship is implemented by placing the primary key of the 1-side as a foreign key in the N-side.
6. An M:N relationship is converted into a separate relation containing the primary keys of both entities.
7. A weak entity contains the primary key of its owner entity as a foreign key.
8. An ISA relationship can be represented using separate tables for specialized entities.
9. Foreign keys maintain referential entities between related tables.

Important constraints

- Primary key
- Foreign key
- NOT NULL
- UNIQUE

- ON DELETE CASCADE
- ON DELETE SET NULL

Relational Schema

The given ER diagram is converted into the following relations:

Customer

```
CUSTOMER(
    Customer_ID PK,
    First_Name,
    Last_Name,
    Email UNIQUE,
    Date_of_Birth,
    Gender
)
```

CUSTOMER_PHONE

```
CUSTOMER_PHONE(
    Customer_ID PK, FK,
    Phone_No PK
)
```

CATEGORY

```
Category(
    Category_ID PK,
    Category_Name UNIQUE,
    Description
)
```

ADDRESS

```
CREATE TABLE Address(
        customer_id INT NOT NULL,
        address_id INT NOT NULL,
        street VARCHAR(100) NOT NULL,
        city VARCHAR(50) NOT NULL,
        state VARCHAR(50) NOT NULL,
        pincode VARCHAR(10) NOT NULL,
        PRIMARY KEY(customer_id, address_id),
        FOREIGN KEY (customer_id) REFERENCES
        CUSTOMER (customer_id) ON DELETE CASCADE

) ENGINE = InnoDB
```

SELLER

```
CREATE TABLE Seller(
        seller_id INT AUTO_INCREMENT PRIMARY KEY,
        seller_name VARCHAR(100) NOT NULL,
        email VARCHAR(100) NOT NULL UNIQUE
)ENGINE = InnoDB;

CREATE TABLE SellerPhone(
        seller_id INT NOT NULL,
        Phone VARCHAR(15) NOT NULL,
        PRIMARY KEY(seller_id, Phone),
        FOREIGN KEY(seller_id) REFERENCES seller(sellerid
                ON DELETE CASCADE

)ENGINE =InnoDB;
```

PRODUCT

```
CREATE TABLE Product(
        product_id INT AUTO_INCREMENT PRIMARY KEY,
        product_name VARCHAR(150) NOT NULL,
        price DECIMAL(10,2) NOT NULL,
        stock INT NOT NULL DEFAULT 0,
        category_id INT NULL,
        seller_id INT NOT NULL,
        FOREIGN KEY(category_id) REFERENCES Category(categoryi
                ON DELETE SET NULL,
        FOREIGN KEY(seller_id) REFERENCES Seller(seller_id)
                ON DELETE CASCADE

) ENGINE =InnoDB;

-- Specialization: Electronics (disjoint subtype of Product)
CREATE TABLE Electronics(
                product_id INT PRIMARY KEY,
                warranty_period INT,
                brand VARCHAR(50)
                FOREIGN KEY(product_id) REFERENCES
                        ON DELETE CASCADE

)ENGINE =InnoDB;
```

-- Specialization : Clothing (disjoint subtype of Product)

```
CREATE TABLE Clothing (
        product_id INT PRIMARY KEY,
        size VARCHAR(10)
        fabric VARCHAR(50)
        FOREIGN KEY (product_id) REFERENCES Product
        (product_id)
                ON DELETE CASCADE
) ENGINE = InnoDB;
```

- ORDER

```
CREATE TABLE Orders(
        order_id INT AUTO_Increment PRIMARY KEY,
        order_date DATE NOT NULL,
        FOREIGN KEY (customer_id) ~~INT NOT NULL~~,
        REFERENCES customer(customer_id)
                ON DELETE CASCADE
) ENGINE = InnoDB;
```

ORDER_ITEM

```
CREATE TABLE Order_Item(
        order_id INT NOT NULL,
        item_no INT NOT NULL,
        quantity INT NOT NULL,
        unit_price DECIMAL(10,2) NOT NULL,
        product_id INT NULL,
        PRIMARY KEY(order_id, item_no)
        FOREIGN KEY (product_id) REFERENCES
        PRODUCT(product_id) ON DELETE SET NULL
) ENGINE = InnoDB ;
```

PAYMENT

```
CREATE TABLE Payment(
        payment_id INT AUTO_INCREMENT PRIMARY KEY
        amount DECIMAL(10,2) NOT NULL,
        payment_date DATE NOT NULL,
        payment_mode VARCHAR(20) NOT NULL,
```

```
    order_id INT NOT NULL UNIQUE,
    FOREIGN KEY(order_id) REFERENCES Orders(order_id)
        ON DELETE CASCADE
)ENGINE=InnoDB;
```

DELIVERY

```
CREATE TABLE DELIVERY(
    delivery_id INT AUTO_INCREMENT PRIMARY KEY,
    amount DECIMAL(10,2) NOT NULL,
    payment_date DATE NOT NULL,
    payment_mode
    Status VARCHAR(20) NOT NULL,
    tracking_number VARCHAR(30)UNIQUE,
    order_id INT NOT NULL UNIQUE,
    FOREIGN KEY(order_id) REFERENCES(order_id)
        ON DELETE CASCADE
)ENGINE=InnoDB;
```

SAMPLE DATA

```
INSERT INTO CUSTOMER(First_name, last_name, email)VALUES
    ('Aarav', 'Sharma', 'aarav.sharma@example.com')
    ('Priya', 'Iyer', 'priya.iyer@example.com');
INSERT INTO Customer_Phone(customer_id, Phone)VALUES
    (1, '9876543210'),
    (2, '9999988888'),
    (2, '9988776655');
INSERT INTO Address(customer_id, address_id, street, city, state,
                    zipcode)VALUES

    (1, 1, '12MG Road', 'Lucknow', 'UttarPradesh', '226012'),
    (2, 1, '45Anna Salai', 'Chennai', 'TamilNadu', '600002');
INSERT INTO CATEGORY(category_name)VALUES
    ('Electronics'), ('Clothing'), ('Groceries');
INSERT INTO seller(seller_name, email) VALUES
    ('Tech Bazar Pvt Ltd', 'contact@techbazar.com'),
    ('Fashionhub India', 'support@fashionhub.in');
```

```
INSERT INTO Seller_Phone (seller_id, Phone) VALUES
(1, '011-40001234'),
(2, '022-50005678');

INSERT INTO PRODUCT (product_name, price, stock, category_id,
seller_id) VALUES
('wireless Earbuds X200', 2499.00, 50, 1, 1),
('Cotton Kurta-Blue', 899.00, 100, 2, 2);

INSERT INTO Orders (~~o~~order_data, total_amount, customer_id)
VALUES
('2026-08-02', 3398.00, 1);

INSERT INTO Electronics (product_id, warranty_period, brand)
~~Voo~~ VALUES
(1, 12, 'Soundwave');

INSERT INTO Order_Item (order_id, item_no, quantity, unit_price,
product_id) VALUES
(1, 1, 1, 2499.00, 1),
(1, 2, 1, 899.00, 2);

INSERT INTO Clothing (product_id, size, fabric) VALUES
~~('2026-08-01',~~
(2, 'M', 'Cotton')

INSERT INTO Payment (amount, payment_date, payment_mode,
order_id) VALUES
(3398.00, '2026-08-01', 'UPI', 1);

INSERT INTO Delivery (delivery_date, status, tracking_number,
order_id) VALUES
('2026-08-05', 'Shipped', 'TRK123456789', 1);
```

Demonstrating Referential Integrity

a) Insert violation: order_id 999 does not exist in orders

Expected: Error 1452 - cannot add or update a child row: a foreign key constraints fails (order_id FK)

~~Insert~~ INSERT INTO Order_Item (Order_id, item_no, quantity, unit_price, product_id)

VALUES(999,1,2,500.00,1);

b) INSERT violation: product_id 999 does not exist in Product
Expected: Error 1452 - foreign key constraint fails (product_id FK)
INSERT INTO Order_Item(order_id, item_no, quantity, unit_price, product_id)
VALUES(1,3,1,10000,999);

c) UNIQUE violation: this order already has a payment (1:1 constraint)
Expected: Error 1062 - Duplicate entry '1' for key 'order_id'
INSERT INTO Payment (amount, payment_date, payment_mode, order_id)
VALUES(500.00,'2026-08-02', 'Cash', 1);

d) ON DELETE SET NULL demonstration: delete a category
Expected: Product.category_id for its products becomes NULL, rows survive
Delete FROM Category WHERE category_id = 1;
SELECT * FROM Product WHERE product_id = 1; category_id is now NULL

e) ON DELETE CASCADE demonstration: delete a Customer
Expected: their Address, Customer_Phone, Orders rows are deleted automatically, which in turn cascades to order_Item, Payment and Delivery for these orders
DELETE FROM Customer WHERE customer_id = 1;
SELECT * FROM Orders ~~where~~ WHERE order_id = 1; empty result
SELECT * FROM Payment WHERE order_id = 1; empty result

f) NOT NULL violation order_date is required
Expected: Error 1364 - Field 'order_date' doesn't have a default value
INSERT INTO orders (total_amount, customer_id) VALUES(500.00,1);